

Dashboard

KnowledgeOS
Enterprise AI Platform

OPERATIONS

Dashboard

Documents

Upload

Connectors

KNOWLEDGE AI

Chunking Studio

Search

Assistant

Comparison

ADMINISTRATION

Settings

Governed Retrieval

Audit-ready pipeline

http://127.0.0.1:5173/dashboard90%

Search documents, connectors, chunks

healthy

KNOWLEDGE OPERATIONS

Enterprise AI control center

New ingestion

Documents16
live backend count

Chunks indexed79
79 embeddings

Vector stores1
Atlas index configured

Connectors2
1 failed jobs

Pipeline Activity

Recent ingestion and retrieval work across enterprise sources.

WORK ITEM	STATUS	OUTPUT	UPDATED
website connector	processed	doc_64dc80a957ff401db4358ce8f7f237c2	19 Jun, 06:05 pm
website connector	failed	website source	19 Jun, 06:05 pm

Quality Gates

Current readiness signals.

Vector index

Ready

RAG grounding

Validated

Secrets exposure

Redacted

Connector retry

Enabled

2. Chunking - used structure based

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http://127.0.0.1:5173/chunking-studio90%

Search documents, connectors, chunks

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CHUNKING STUDIO

Prepare retrieval-ready context

Apply strategy

Strategy

Preview before persisting chunks.

Document

RAG_Chunking_Study_Guide.pdf

Method

Structure-Based

Chunk size

500

Overlap

50

☒ Replace existing chunks for this strategy

{}

Preview chunks

Preview

Chunk boundaries, tokens, and metadata.

chunk_001

Retrieval-Augmented Generation (RAG) Chunking

4 tokens | structure_based | ("chunk_index":0,"start_token":0,"end_token":4,"section_index":0,"split_method":"structure_based")

chunk_002

Strategies A Comprehensive Study & Decision Guide for Document Segmentation Core Concept: In RAG systems, large documents must be split into smaller blocks (chunks) before being converted into vector embeddings and indexed. The choice of chunking strategy directly determines the performance, accuracy, and retrieval cost of your Large Language Model (LLM) pipeline. A poor strategy results in lost context, fragmented meanings, and bloated token costs, while an optimized strategy retrieves highly accurate, contextual answers.

74 tokens | structure_based | ("chunk_index":1,"start_token":0,"end_token":74,"section_index":1,"split_method":"structure_based")

chunk_003

Why is Chunking Critical in RAG? 1. Context Window Limits: LLMs have physical constraints on input sizes. Proper chunking fits content cleanly. 2. Search Relevance: Vector databases retrieve matching segments based on semantic similarity. Smaller, coherent chunks ensure the database retrieves only relevant information, ignoring irrelevant noise. 3. Token Economy: Injecting entire multi-page manuals for simple queries is highly expensive and slow. Chunking minimizes LLM generation fees. 4. Semantic Continuity: Sentences and tables must be kept intact so their logical relationships are not severed.

84 tokens | structure_based | ("chunk_index":2,"start_token":0,"end_token":84,"section_index":2,"split_method":"structure_based")

chunk_004

Detailed Chunking Strategies

KnowledgeOS
Enterprise AI Platform

OPERATIONS

Dashboard

Documents

Upload

Connectors

KNOWLEDGE AIDivChunking Studio

Search

Assistant

Comparison

ADMINISTRATION

Settings

Governed Retrieval
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http://127.0.0.1:5173/chunking-studio

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CHUNKING STUDIO

Apply strategy

Strategy

Preview before persisting chunks.

Document

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Method

Structure-Based

Chunk size

500

Overlap

50

☒ Replace existing chunks for this strategy

{ } Preview chunks

153 chunks applied
153 embeddings created
processed

Preview

Chunk boundaries, tokens, and metadata.

chunk_001
Retrieval-Augmented Generation (RAG) Chunking
4 tokens | structure_based | {"chunk_index":0,"start_token":0,"end_token":4,"section_index":0,"split_method":"structure_based"}

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chunk_004
Detailed Chunking Strategies

3. Retrieval

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Enterprise AI Platform

OPERATIONS

Dashboard

Documents

Upload

Connectors

KNOWLEDGE AIDivChunking Studio

Search

Assistant

Comparison

ADMINISTRATION

Settings

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Audit-ready pipeline

http://127.0.0.1:5173/search

90%

healthy

Search documents, connectors, chunks

Why is chunking required

hybridvectorkeywordagenticAll sources

5

Run

Results

5 results using vector

RAG_Chunking_Study_Guide.pdf
Score 0.843116 | vector | doc_b69879a5a14440fb910588e33b6400c2
Detailed Chunking Strategies

RAG_Chunking_Study_Guide.pdf
Score 0.829151 | vector | doc_b69879a5a14440fb910588e33b6400c2
[Chunk 2]: 'necessary. It prevents excee'

RAG_Chunking_Study_Guide.pdf
Score 0.818003 | vector | doc_b69879a5a14440fb910588e33b6400c2
Why is Chunking Critical in RAG? 1. Context Window Limits: LLMs have physical constraints on input sizes. Proper chunking fits content cleanly. 2. Search Relevance: Vector databases retrieve matching segments based on semantic similarity...

RAG_Chunking_Study_Guide.pdf
Score 0.802317 | vector | doc_b69879a5a14440fb910588e33b6400c2
[Chunk 1]: 'In RAG, chunking is necessary.'

RAG_Chunking_Study_Guide.pdf
Score 0.786127 | vector | doc_b69879a5a14440fb910588e33b6400c2
3. Document-Structure Aware Chunking Overview: Leverages formatting elements (like Markdown headings H1-H4, HTML section blocks, or PDF chapters) to segment the document. Each heading naturally marks the boundaries of a new chunk, preser...

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KNOWLEDGE AI

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RETRIEVAL

Search knowledge index

Q Why is chunking required

hybrid vector keyword agentic All sources

5

Run

Results

5 results using hybrid

RAG_Chunking_Study_Guide.pdf

Score 0.890802 | hybrid | doc_b69879a5a14440fb910588e33b6400c2

Why is Chunking Critical in RAG? 1. Context Window Limits: LLMs have physical constraints on input sizes. Proper chunking fits content cleanly. 2. Search Relevance: Vector databases retrieve matching segments based on semantic similarity...

RAG_Chunking_Study_Guide.pdf

Score 0.60587 | hybrid | doc_b69879a5a14440fb910588e33b6400c2

Detailed Chunking Strategies

RAG_Chunking_Study_Guide.pdf

Score 0.497491 | hybrid | doc_b69879a5a14440fb910588e33b6400c2

[Chunk 2]: 'necessary. It prevents excee'

RAG_Chunking_Study_Guide.pdf

Score 0.48139 | hybrid | doc_b69879a5a14440fb910588e33b6400c2

[Chunk 1]: 'In RAG, chunking is necessary.'

RAG_Chunking_Study_Guide.pdf

Score 0.471676 | hybrid | doc_b69879a5a14440fb910588e33b6400c2

4. Retrieval with LLM

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RAG ASSISTANT

Grounded enterprise answers

Chunking is required for several reasons: 1. **Context Window Limits**: LLMs have physical constraints on input sizes, and proper chunking ensures that content fits cleanly within these limits [S3]. 2. **Search Relevance**: Vector databases retrieve matching segments based on semantic similarity, making chunking necessary for effective search and retrieval [S3]. 3. **Prevents excessive input**: Chunking prevents excessive input sizes, which is mentioned as a necessary aspect of the process [S2]. Overall, chunking is critical in RAG (Retrieval-Augmented Generation) systems, as mentioned in [S4] and [S3]. References: [S2] [S3] [S4]

healthy Confidence 0.8301 5 sources vector

why is chunking required

Grounding Controls

Retrieval settings for governed responses.

Top K 5

Strategy ☒ Agentic

Source citations **Required**

Sources and Retrieved Chunks

Grounding evidence returned by the backend.

RAG_Chunking_Study_Guide.pdf

Detailed Chunking Strategies